

POWER BI - WEEK 3: FINANCIAL DATA IMPORT, TRANSFORMATION & TIME INTELLIGENCE

Name: Durga Devi S

Roll No: E25DA018

Programme: B.Sc. Data Science and
Artificial Intelligence

INTRODUCTION:

This week focuses on analyzing **Shopify stock market data** using Power BI. The dataset contains historical information about stock prices and trading volume, including Open, High, Low, Close, Adjusted Close, and Volume values. The data will be imported and prepared using Power Query, followed by the creation of a dedicated Date Dimension using DAX. Financial measures and time-intelligence calculations will then be developed to analyze stock performance over time. Finally, a structured data model with date hierarchies and an active relationship will be created to support effective financial analysis and drill-down reporting.

Figure 1: Shopify stock market dataset imported into Power Query Editor for financial time-series analysis

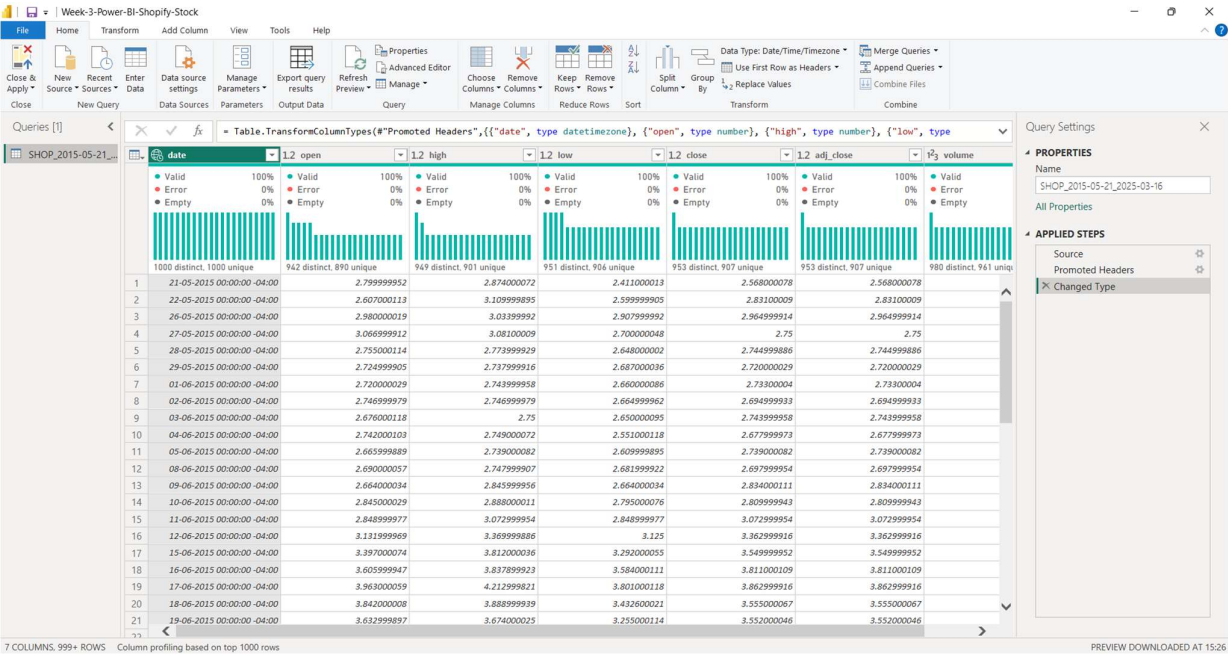


Figure 2: Shopify stock data cleaned and validated with appropriate data types in Power Query

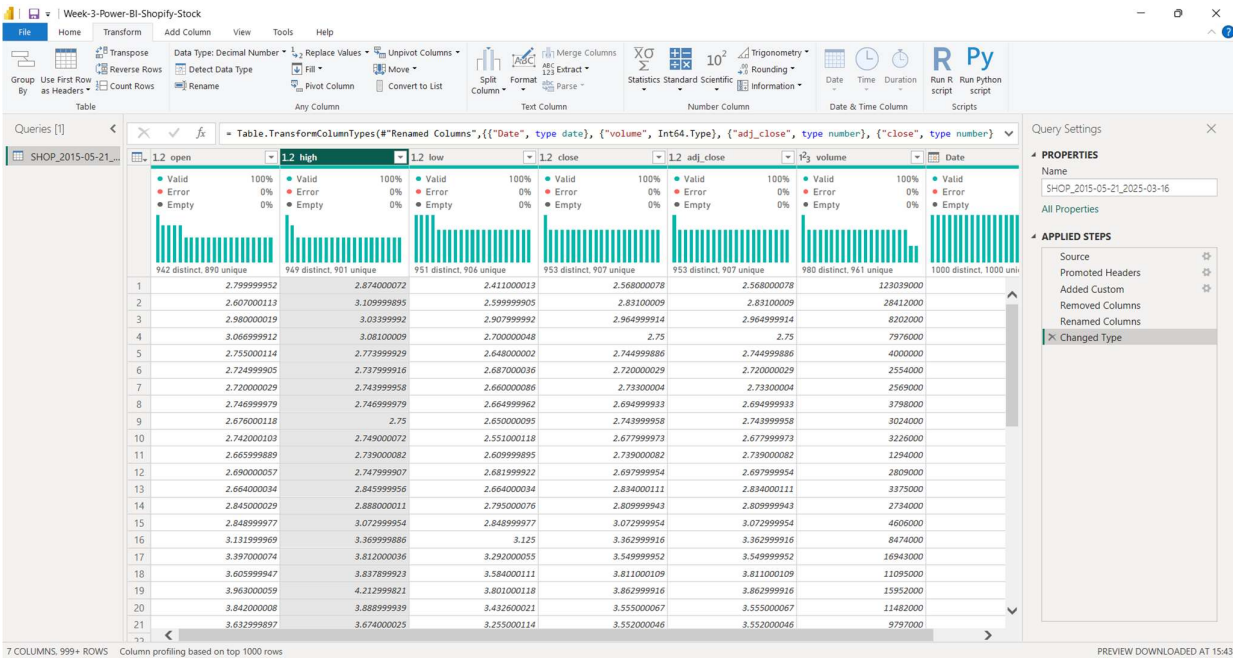


Figure 3: Dedicated Dim_Date table generated using the DAX CALENDARAUTO function

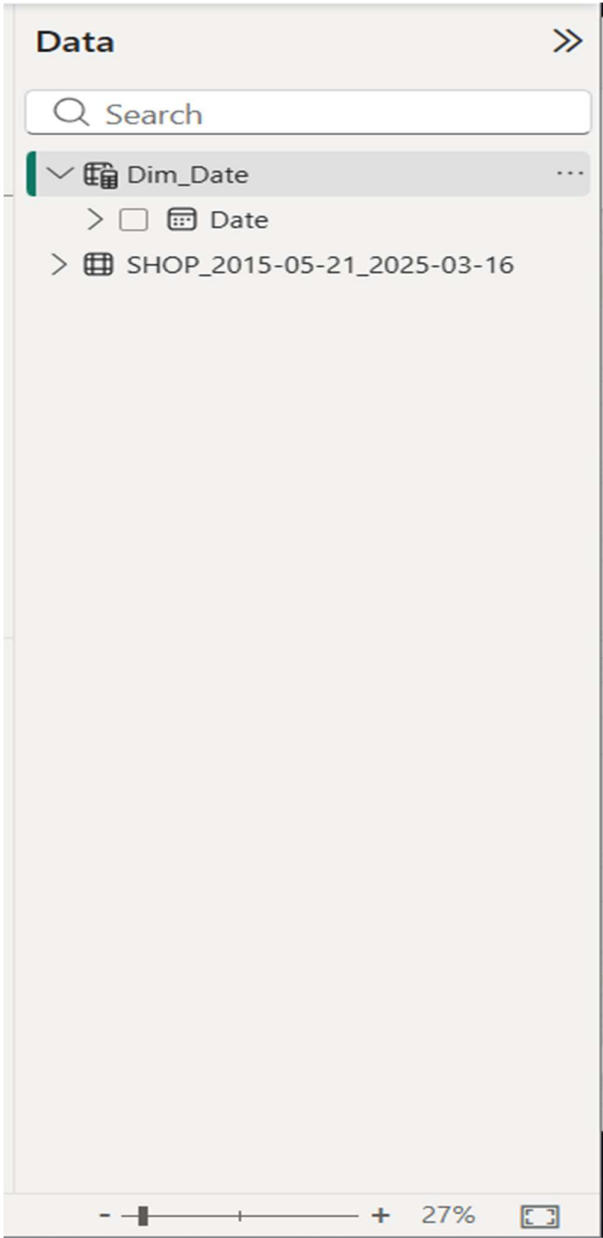


Figure 4: Custom date attributes created in Dim_Date for time-based financial analysis and drill-down

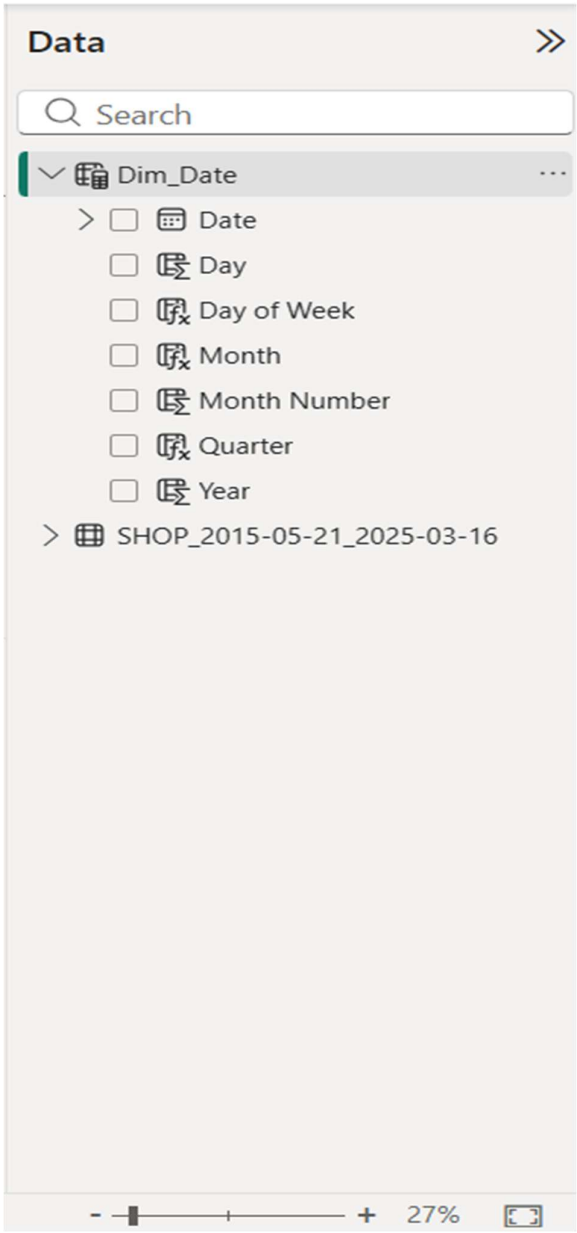


Figure 5: Dim_Date marked as the official Date Table with the Date column validated successfully

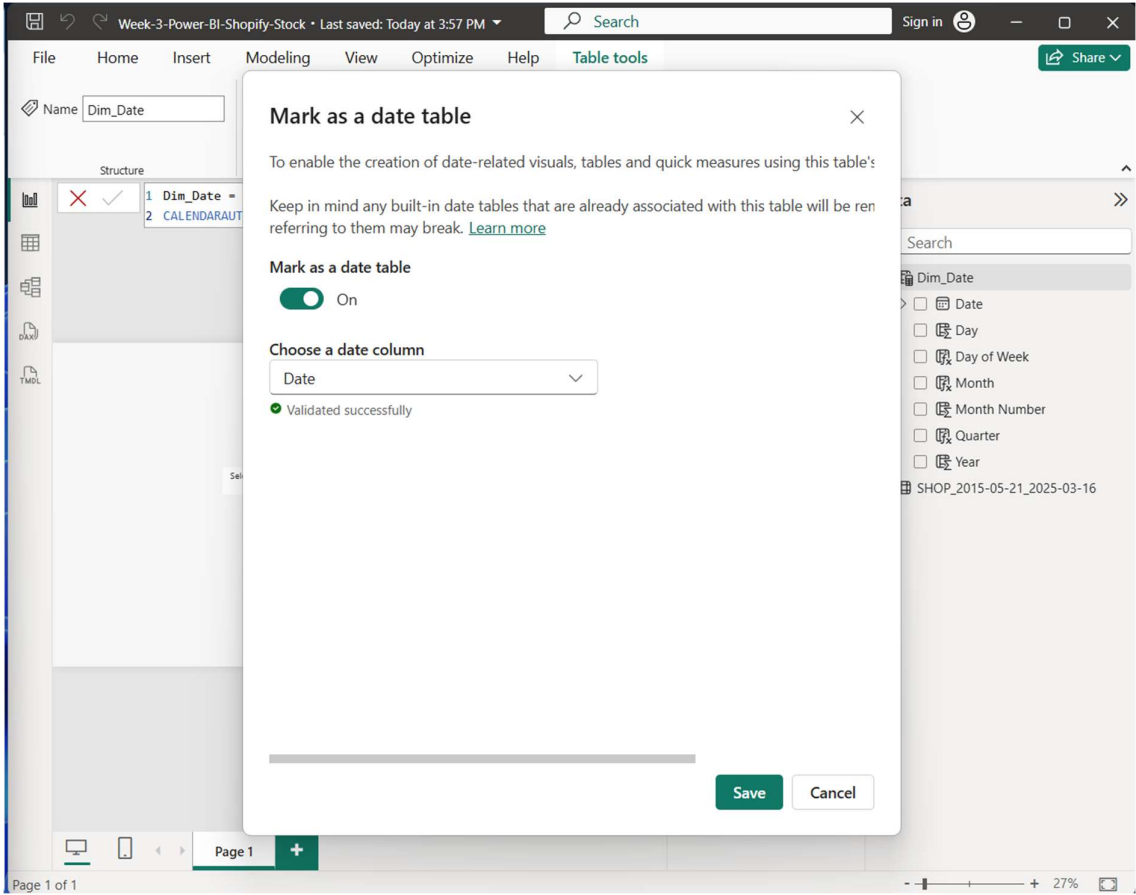


Figure 6: Active one-to-many relationship established between Dim_Date and Shopify stock price data

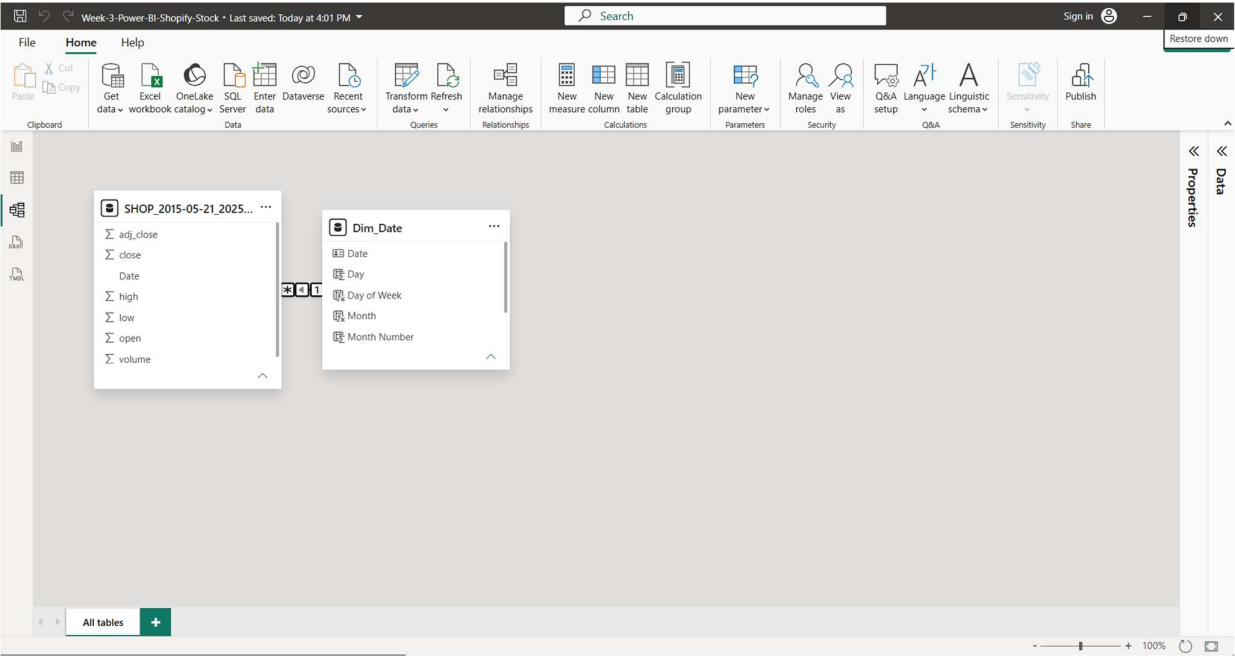


Figure 7: Core financial DAX measures created for Shopify stock price and trading activity analysis

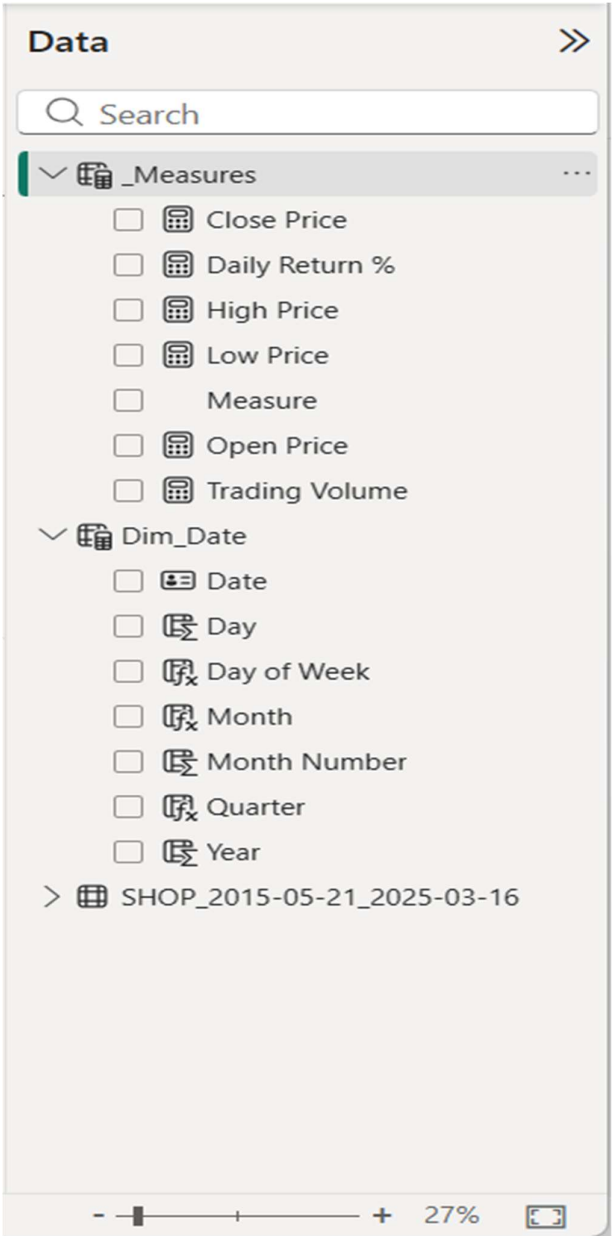


Figure 8: Time-intelligence DAX measures created using DATEADD, TOTALYTD, and SAMEPERIODLASTYEAR

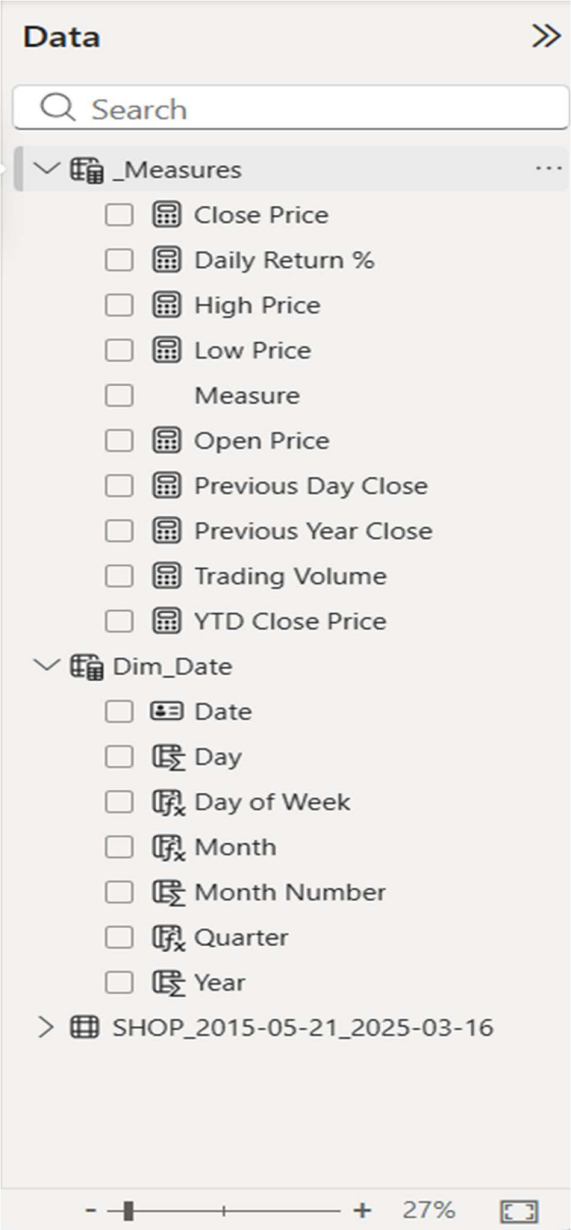


Figure 9: Custom Year-Quarter-Month-Day hierarchy created in Dim_Date for time-based drill-down analysis

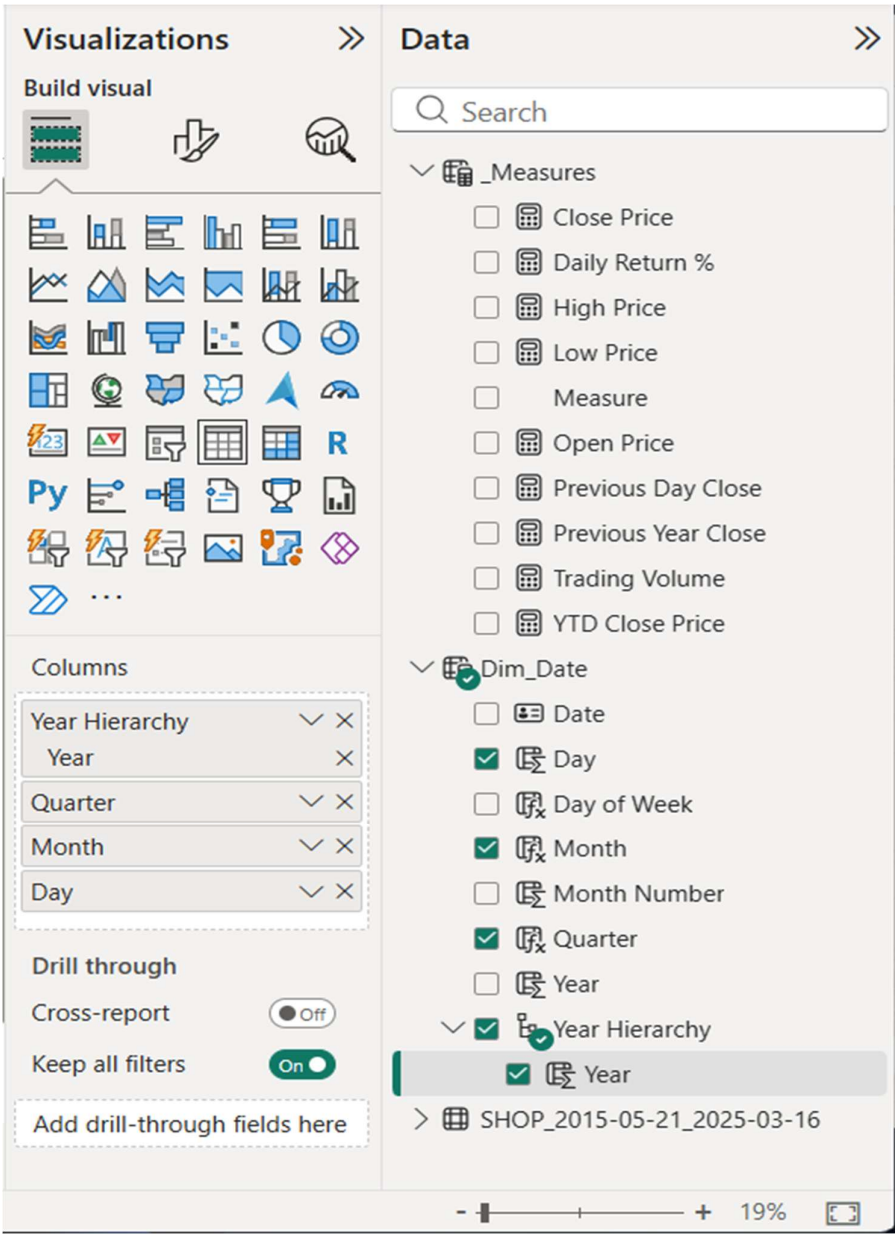


Figure 10: Final structured Power BI data model for Shopify financial analytics

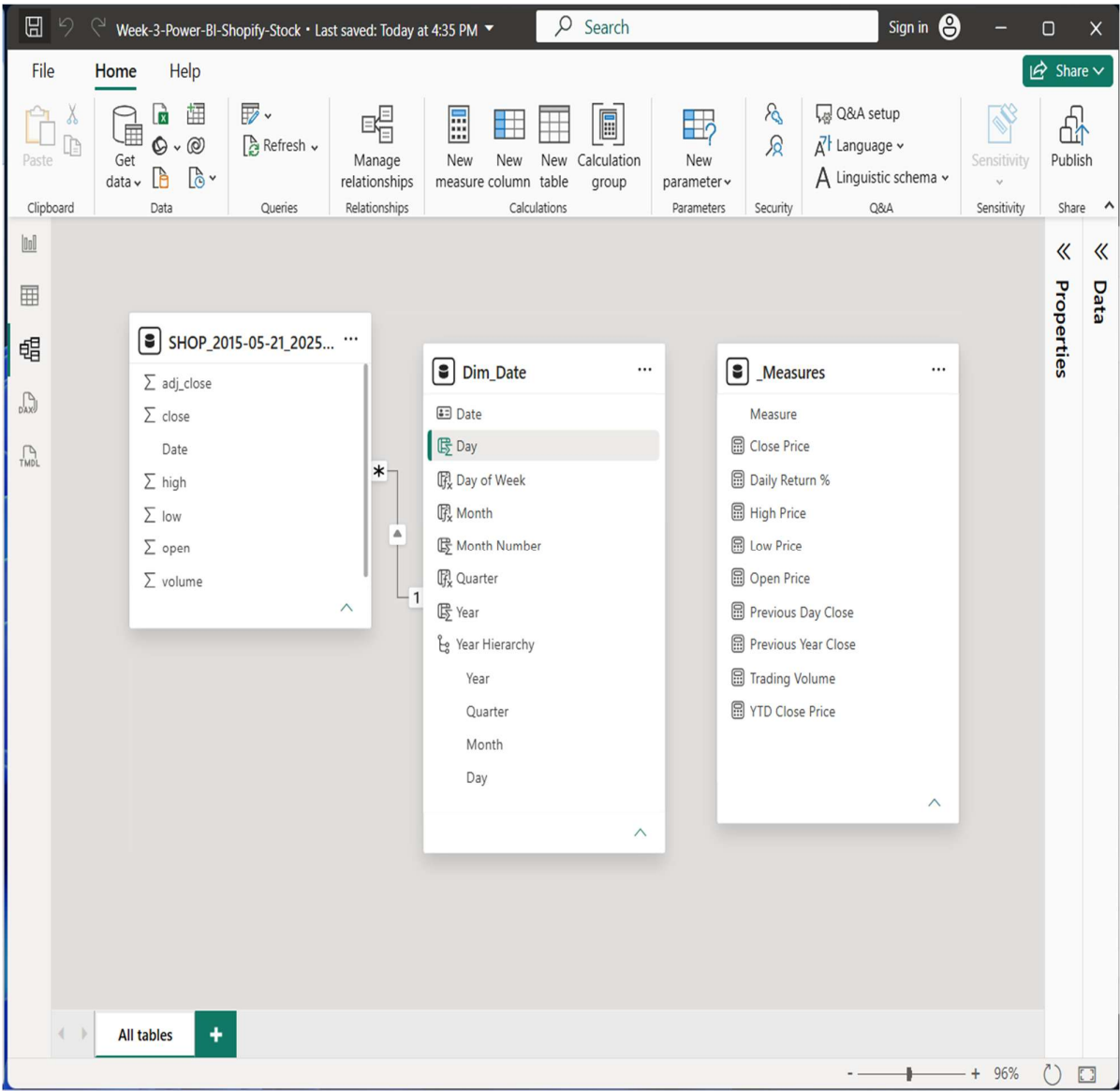


Figure 11: Financial KPI cards displaying key Shopify stock price and trading activity measures

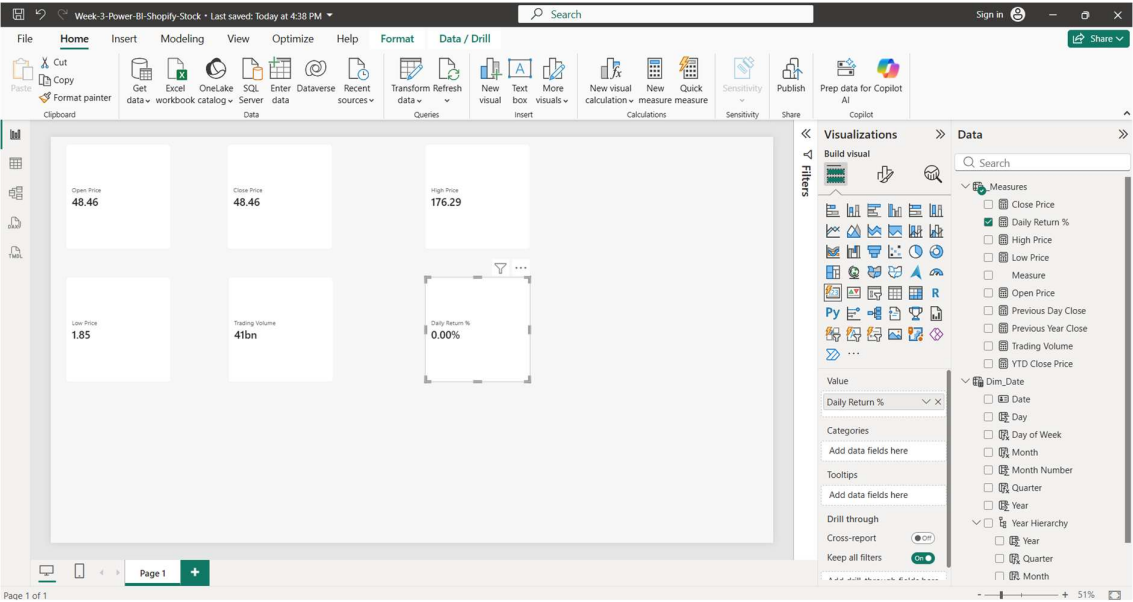


Figure 12: Shopify closing price trend over time using the Dim_Date time hierarchy

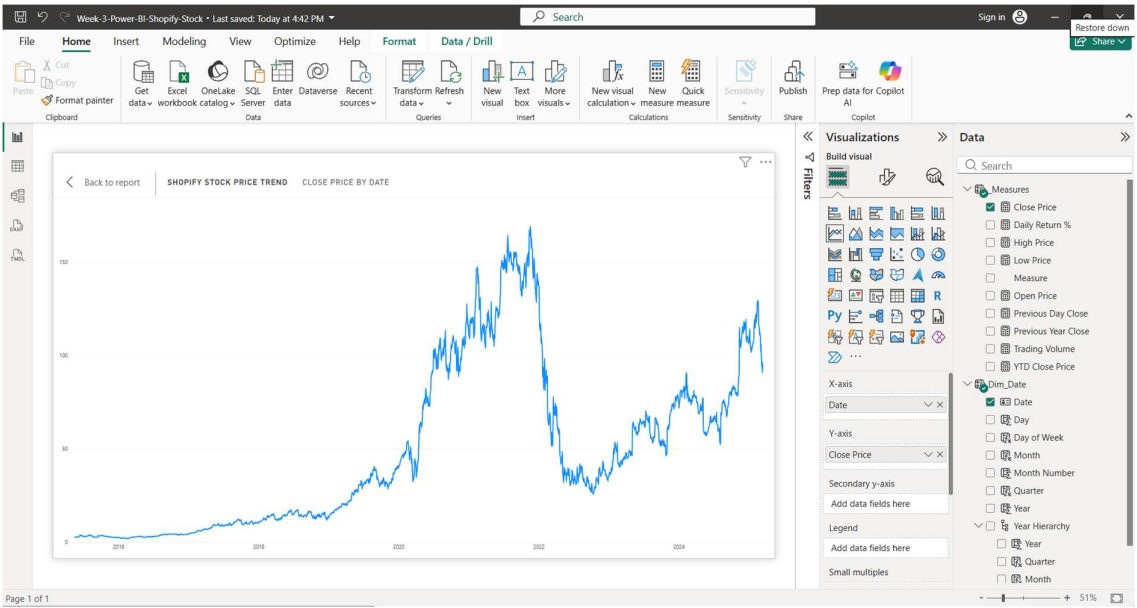


Figure 13: Shopify trading volume analyzed over time using the dedicated Date Dimension

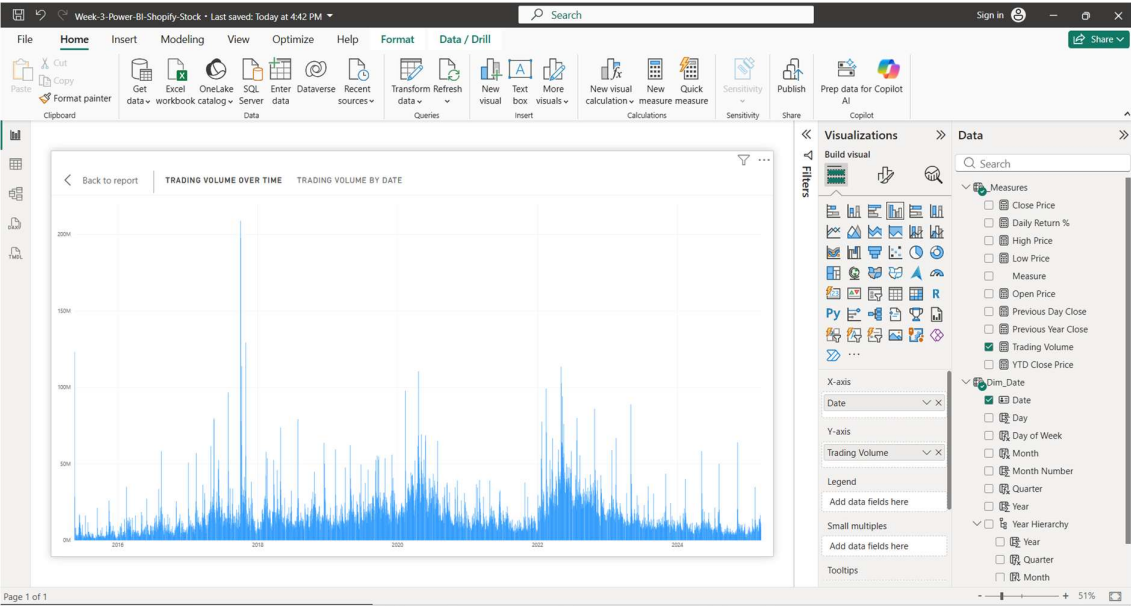


Figure 14: Daily percentage return trend of Shopify stock over time

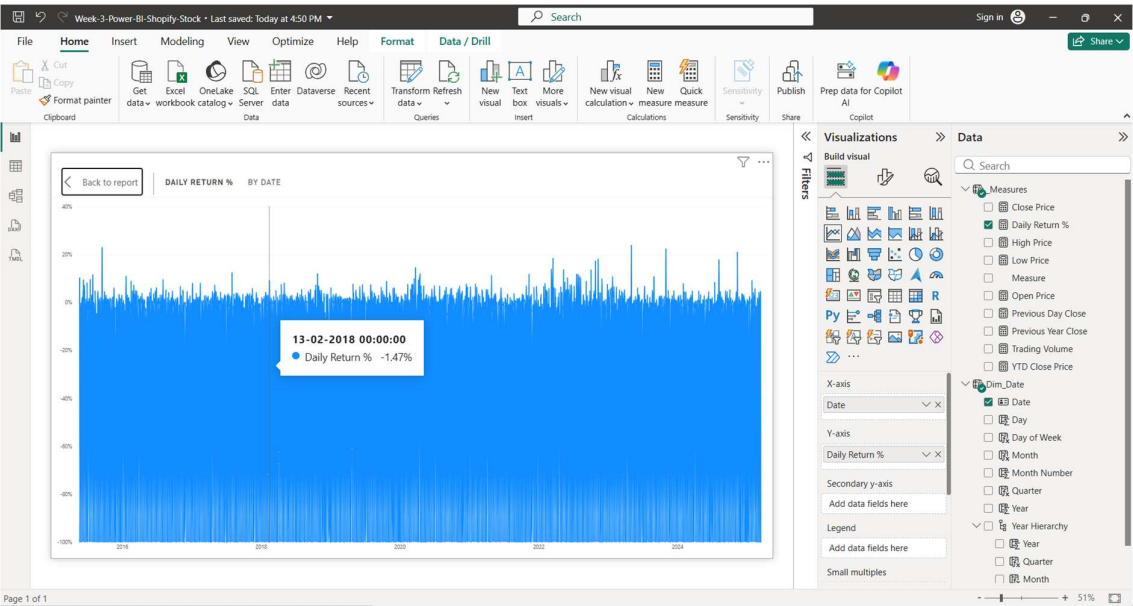
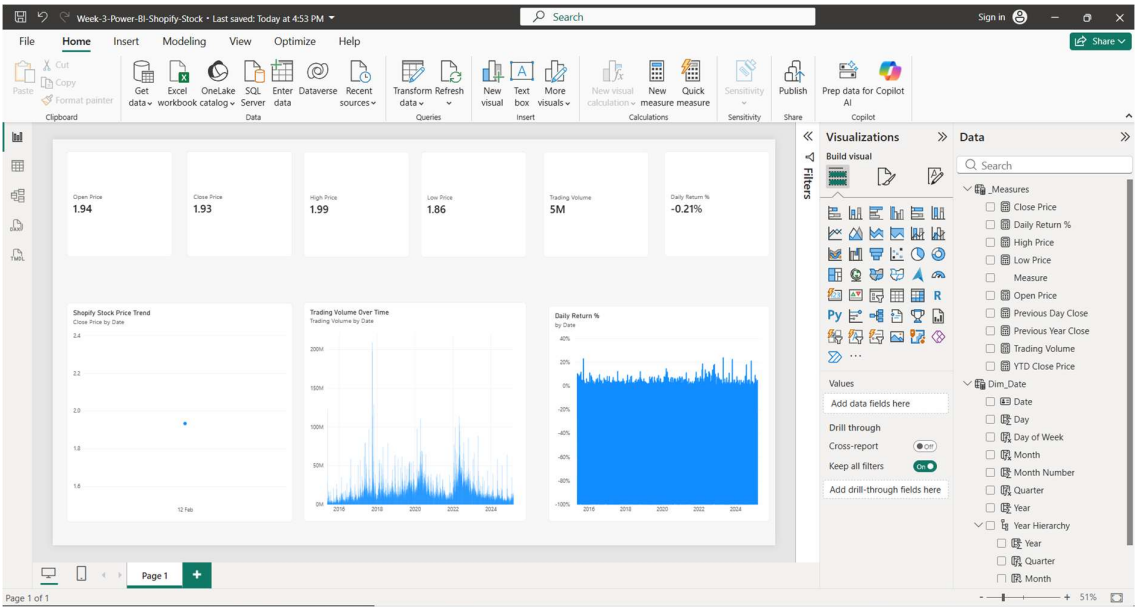


Figure 15: Final interactive Shopify Stock Financial Analysis Dashboard



Conclusion

The Shopify stock market dataset was successfully imported and transformed using Power Query in Power BI. A dedicated Dim_Date table was created using CALENDARAUTO() and marked as the official Date Table. An active one-to-many relationship was established between the Date Dimension and the Shopify stock data. Core financial DAX measures were developed for Open Price, Close Price, High Price, Low Price, Trading Volume, and Daily Return %. Time-intelligence calculations using DATEADD, TOTALYTD, and SAMEPERIODLASTYEAR were also implemented. A Year-Quarter-Month-Day hierarchy was created to support drill-down analysis. Finally, an interactive financial dashboard was developed to analyze Shopify stock prices, trading volume, and daily returns over time.